



NEXT CHAPTER @ SMITH PUBLIC LIBRARY

Meet the Mystery Box

Session 4 — Basic Algebra: Expressions & Solving for x

Monday Evenings • 5:30–7:30 PM • Library Conference Room

Tonight's Agenda

1

Welcome & quick recap

10 min

2

What is a variable? Expressions vs. equations

15 min

3

One-step equations: concept, guided practice, answers

25 min

4

Two-step equations: concept, guided practice, answers

35 min

5

Worked real-world examples together

20 min

6

Wrap-up & take-home preview

15 min

Quick Recap: Last Week

Ratios:

Find the scale factor, apply it to both sides.

Percentage applications:

Discounts, tips, and tax all use the same move — multiply the percentage (as a decimal) by the starting amount.

That completes Math Part A and B. Tonight we start something new: algebra.



What Is a Variable?

A variable (like x) is just a box holding a number you don't know yet.

It's not scary math magic — it's a placeholder, like a blank on a form waiting to be filled in.

“Solving for x ” just means figuring out what number the box is hiding.

Expressions vs. Equations

An Expression

A math phrase with no equals sign. No single right answer — just a description.

$x + 5$ “some number, plus 5”

$3x$ “three times some number”

An Equation

A math sentence WITH an equals sign. It says two things are equal — and it can be solved.

$x + 5 = 12$ solve: $x = 7$

$3x = 21$ solve: $x = 7$



Solving One-Step Equations

The Balance Rule:

whatever you do to one side, you must do to the other side too.

Addition/Subtraction: $x + 5 = 12 \rightarrow$ subtract 5 from both sides $\rightarrow x = 7$

Multiplication/Division: $3x = 21 \rightarrow$ divide both sides by 3 $\rightarrow x = 7$

Your Turn: One-Step Equations

GUIDED PRACTICE

1

$$x + 6 = 14$$

2

$$x - 3 = 9$$

3

$$4x = 20$$

4

$$x \div 2 = 8$$

Let's Check Your Answers



1. $x + 6 = 14 \rightarrow \text{subtract } 6 \rightarrow x = 8$



2. $x - 3 = 9 \rightarrow \text{add } 3 \rightarrow x = 12$



3. $4x = 20 \rightarrow \text{divide by } 4 \rightarrow x = 5$



4. $x \div 2 = 8 \rightarrow \text{multiply by } 2 \rightarrow x = 16$

Solving Two-Step Equations

Same balance rule — just two moves instead of one. Undo addition/subtraction first, then multiplication/division.

Example: $2x + 5 = 15$

Step 1: Subtract 5 from both sides $\rightarrow 2x = 10$

Step 2: Divide both sides by 2 $\rightarrow x = 5$

Your Turn: Two-Step Equations

GUIDED PRACTICE

1

$$3x + 4 = 19$$

2

$$2x - 6 = 10$$

3

$$5x + 1 = 21$$

4

$$4x - 3 = 13$$

Let's Check Your Answers



1. $3x + 4 = 19 \rightarrow$ subtract 4 $\rightarrow 3x = 15 \rightarrow$ divide by 3 $\rightarrow x = 5$



2. $2x - 6 = 10 \rightarrow$ add 6 $\rightarrow 2x = 16 \rightarrow$ divide by 2 $\rightarrow x = 8$



3. $5x + 1 = 21 \rightarrow$ subtract 1 $\rightarrow 5x = 20 \rightarrow$ divide by 5 $\rightarrow x = 4$



4. $4x - 3 = 13 \rightarrow$ add 3 $\rightarrow 4x = 16 \rightarrow$ divide by 4 $\rightarrow x = 4$

Let's Solve One Together

You earn \$15 an hour, plus a one-time \$20 bonus. Your paycheck this week is \$110. How many hours did you work?

Set it up: $15x + 20 = 110$

Step 1: Subtract 20 from both sides $\rightarrow 15x = 90$

Step 2: Divide both sides by 15 $\rightarrow x = 6$

You worked 6 hours.

Before You Go



Try tonight's take-home practice

Real practice questions on tonight's material — 10 minutes, any time this week.



Thursday office hours are open

If “x” still feels unfamiliar, this is the place to ask — bring the Extra Help handout if you used it.



Next week: more Math

Linear equations and graphing — building directly on tonight's balance rule.



You Just Solved for x .

See you next Monday — linear equations and graphing.

Before next week:

Try tonight's take-home practice. Want more? GED.com's free test preview tutorials (tools.ged.com) are always open, in English or Spanish.